A Survey of LLM x Data

Challenges and Future Directions in Data and LLM Systems

M. Huseynzade

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Outline

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Introduction

Chapter 4 Overview

- **Theme**: Challenges and future directions at the intersection of Data Management and LLMs.
- Two main perspectives:
  1. **Data Management for LLM** – How to manage data for LLM training and serving.
  2. **LLM for Data Management** – How to use LLMs to improve data systems.
- Focus: Open problems and research opportunities.

Data Management for LLM

Key Challenges in Data for LLMs

1. Task-Specific Data Selection

- General pretraining data often irrelevant to target tasks.
- Need dynamic selection methods, not just fixed rules.

2. Optimizing Data Processing Pipelines

- Current pipelines rely on costly trial-and-error.
- Goal: Predict optimal preprocessing configurations without full training.

More Challenges in Data for LLMs

3. Knowledge Update and Version Control

- Fast-changing domains (health, finance) need up-to-date knowledge.
- Need platforms for dataset sharing, versioning, and real-time updates.

4. Comprehensive Dataset Evaluation

- Hard to judge dataset quality without training.
- Need model-agnostic evaluation metrics that correlate with performance.

Hybrid RAG Indexing and Retrieval

- Current systems are siloed: Elasticsearch (text/vector), LightRAG (vector/graph).
- **Future goal**: Unified indexing engine supporting:
  - Full-text search
  - Vector similarity
  - Knowledge graphs
  - Structured queries
- Platforms like AutoRAG exist but focus on pipelines, not integrated indexing.

LLM for Data Management

Unified Data Analysis System

- Current: Separate models for different data types (structured, unstructured, etc.).
- High deployment/maintenance cost.
- **Future**: Single model handling all data types.
- Goal: Generalized data analysis capability.

Data Analysis with Private Domain Knowledge

- Current: RAG or fine-tuning, but struggles with novel/complex domains.
- Example: Text-to-SQL on DB with 10K columns + domain knowledge per column.
- Lack of datasets with explicit domain knowledge.
- **Open problem**: Better integration of domain knowledge into LLMs.

Representing Non-Textual Data

- Current: Flatten structured data (e.g., tables) into text.
- Loss of structural semantics (row/column relationships).
- Statistical signals (selectivities, histograms) poorly represented.
- **Future**:
  - Multi-modal LLMs
  - Tailored encoders preserving structure

Efficient LLM Usage Under Budget Constraints

- LLMs are expensive and slow for large-scale tasks.
- Examples:
  - Table analysis on millions of rows
  - Query optimization taking minutes vs. milliseconds
- **Future**: Hybrid strategies:
  - Combine LLMs with traditional methods
  - Smart scheduling across multiple LLMs
  - Cost-performance trade-offs

Summary

Key Takeaways

1. **Data for LLMs**: Need smarter data selection, evaluation, updating, and hybrid indexing.
2. **LLMs for Data**: Need unified systems, better domain knowledge integration, richer data representations, and cost-efficient deployment.
3. Both directions require integrated, scalable, and intelligent systems.

Future Vision

Toward Intelligent, Unified, and Efficient

Data–LLM Ecosystems

- Data systems that understand and serve LLMs
- LLMs that understand and optimize data systems
- Hybrid approaches for real-world scalability

Paper: DataAI-2025 Survey
Chapter 4: Challenges and Future Directions

Thank You!

Questions?

Presentation based on:

DataAI-2025 Survey – Chapter 4

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